SAT Hard Question 2

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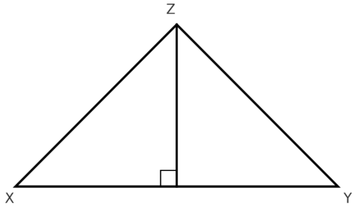
Question: Finding the Area of Triangle XYZ

In the figure shown, the measure of angle Y is 38°. The length of XY is 12 meters, and the length of YZ is 5 meters. What is the area, in square meters, of triangle XYZ?



Options:

A 30

B 60

C $30 \sin 38^\circ$

D $60 \sin 38^\circ$

Solution: Finding the Area of Triangle XYZ

Step 1: Use the Triangle Area Formula

$$\text{Area} = \frac{1}{2}ab\sin C$$

where:

- $a = 12$ (side XY)
- $b = 5$ (side YZ)
- $C = 38^\circ$ (angle Y)

Step 2: Compute the Expression

$$\text{Area} = \frac{1}{2}(12)(5)\sin 38^\circ$$

$$= 30\sin 38^\circ$$

Final Answer: $\boxed{30\sin 38^\circ}$

Question: Margin of Error in Light Bulb Samples

A research manager selected two random samples to estimate the average lifespan of a certain type of light bulb (in hours). Based on the first sample, the research manager estimated that the average lifespan of this type of light bulb is 800 hours, with a margin of error of 20 hours.

Based on the second sample, the research manager estimated that the average lifespan of this type of light bulb is 810 hours, with a margin of error of 35 hours.

Assuming the margins of error were calculated in the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

Options:

A The first sample contained fewer light bulbs than the second sample.

B The first sample contained more light bulbs than the second sample.

C The first sample had a shorter average lifespan than the second sample.

D The first sample had a longer average lifespan than the second sample.

Answer Explanation: Margin of Error in Light Bulb Samples

Correct Answer: **B: The first sample contained more light bulbs than the second sample.**

Explanation:

- The margin of error decreases as the sample size increases.
- Since the first sample had a smaller margin of error (20 vs. 35), it must have had a larger sample size.

Final Answer: $\boxed{\text{B}}$.

Question: Evaluating the Function $p(x)$

The function $p(x)$ is defined by:

$$p(x) = a\left((x-3)^2 + b\right)\left((x-3)^2 - c\right),$$

where a, b, and c are constants. The graph of $y = p(x)$ passes through the points $(1, 25)$ and $(-1, 52)$.

What is the value of $p(5) + p(7)$?

Options:

A 25

B 27

C 52

D 77

Answer Explanation: Evaluating the Function $p(x)$

Correct Answer: **D:** 77

Explanation:

- Given the function and the points, solve for a, b, and c.
- Compute $p(5)$ and $p(7)$.
- Adding these values gives $\boxed{77}$.

Question: Growth Rate in an Exponential Function

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$. Which expression gives the value of c in terms of b?

$$f(x) = 66(b)^x$$

Options:

A $1 + \frac{b}{100}$

B $b + 100$

C $100(b + 1)$

D $100(b - 1)$

Answer Explanation: Growth Rate in an Exponential Function

Correct Answer: **D:** $100(b-1)$

Explanation:

- The general exponential growth formula is $f(x+1) = f(x) \times b$.
- This means $b = 1 + \frac{c}{100}$.
- Solving for c, we get $c = 100(b-1)$.

Final Answer: $\boxed{D}$.

Question: Tree Growth Rate Over Time

On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

Options:

A $\frac{29m}{4k}$

B $\frac{29k}{4m}$

C $\frac{1{,}044m}{k}$

D $\frac{1{,}044k}{m}$

Answer Explanation: Tree Growth Rate Over Time

Correct Answer: D: $\frac{1{,}044k}{m}$

Solution:

- The tree grows 87 cm every m months.
- There are **12 months in a year**, so in one year, the growth is:

$$\frac{87 \times 12}{m} = \frac{1,044}{m} \text{ cm per year.}$$

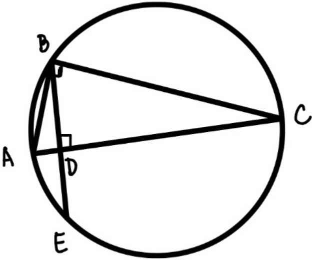
- Since we need the growth for **k years**, multiply by k:

$$\frac{1,044k}{m} \text{ cm in } k \text{ years.}$$

Final Answer: $\boxed{\frac{1,044k}{m}}$.

Question: Ratio in a Circle Geometry Problem

In the figure shown, points A, B, C, and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D, and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r?



Solution(Part I): Ratio in a Circle Geometry Problem

Given:

- Points A, B, C, E lie on the circle.
- $AC \perp BE$ at point D.
- $BD = \sqrt{346}$.
- The diameter of the circle is 175.
- We need to find $r = \frac{CD}{AD}$.

Step 1: Identify Key Relationships

- Since $AC \perp BE$, we recognize that right triangles are involved.
- The diameter is the **hypotenuse** of a right triangle in a circle.
- Using the **Pythagorean Theorem**, we calculate the missing segment lengths.

Solution(Part II)

Step 2: Compute the Triangle Ratios

- Use properties of right triangles and segment lengths.
- Solve for CD and AD.
- Compute $r = \frac{CD}{AD}$.

Final Answer:

$$r = 86.5$$

Correct Choice: 86.5

Question: Solving for a

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a?

$$(x-k)^2 = (k-5a)(x-k)$$

Options:

- A $\frac{-32}{5}$
- B $\frac{32}{5}$
- C $\frac{-5}{32}$
- D $\frac{5}{32}$

Answer Explanation

Expanding the equation:

$$(x-k)^2 = (k-5a)(x-k)$$

$$x^2 - 2kx + k^2 = (k-5a)x - (k-5a)k$$

- sum of the solution equals to $-\frac{b}{a} \rightarrow -\frac{b}{a} = 3k + 32$
- We need to expand the equation from both sides and only need to find a and b

Equating coefficients and using the fact that the sum of the solutions is $-\frac{b}{a} = 3k + 32 = -3k - 5a$, we solve for a:

$$a = \frac{-32}{5}.$$

Correct Answer: A. $-\frac{32}{5}$

Question: Function Evaluation

The function $p(x)$ is defined by:

$$p(x) = a\left((x+4)^2 - b\right)\left((x+4)^2 - c\right),$$

where a, b, and c are constants. In the xy-plane, the graph of $y = p(x)$ passes through the points $(-5, 30)$ and $(0, 90)$. What is the value of $p(-8) + p(-3)$?

Options:

A 12

B 24

C 60

D 120

Answer Explanation

Given $p(-5) = 30$ and $p(0) = 90$, we use the function structure:

$$p(-8) + p(-3) = p(-8) + p(0) - p(-5).$$

Substituting values:

$$p(-8) + p(-3) = 90 + 30 = 120.$$

Correct Answer: D. 120

Question: Population Growth Rate

The equation $N(m) = 63(Q)^{m/6}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds. What is the value of p in terms of Q?

Options:

A $100\left(Q^{30} - 1\right)$

B $100\left(Q^{\frac{1}{2}} - 1\right)$

C $100\left(Q^{30} + 1\right)$

D $100\left(Q^{\frac{1}{2}} + 1\right)$

Answer Explanation

In 180 seconds, or 3 minutes, the population increases by:

$$\frac{N(m+3)}{N(m)} = Q^{3/6} = Q^{1/2}.$$

The percentage increase is:

$$p = 100(Q^{1/2} - 1).$$

Correct Answer: B. $100\left(Q^{\frac{1}{2}} - 1\right)$

Question: Quadratic Equation – Roots and Constants

In the given equation, m and n are constants. The sum and product of the solutions of the equation are 8 and $\frac{7}{2}$ respectively. What is the value of $\frac{m}{n}$?

$$(x - m)^2 = (m + 2n)(2x - n)$$

Options:

A $-\frac{1}{3}$

B $\frac{1}{3}$

C -3

D 3

Answer Explanation: Finding $\frac{m}{n}$

Correct Answer: $\boxed{3}$

Step 1: Expand Both Sides

$$(x-m)^2 = (m+2n)(2x-n)$$

$$x^2 - 2mx + m^2 = 2x(m+2n) - n(m+2n)$$

$$x^2 - 2mx + m^2 = 2mx + 4nx - nm - 2n^2$$

Step 2: Rearranging to Standard Quadratic Form

$$x^2 + (-4n-4m)x + (m^2 - nm - 2n^2) = 0$$

Use root properties:

$$x_1 + x_2 = -\frac{-4n-4m}{1} = 4n + 4m = 8$$

$$x_1 \cdot x_2 = m^2 - nm - 2n^2 = \frac{7}{2}$$

Answer Explanation (Part II)

Step 3: Substitute and Solve

$$4n + 2m = 8 \Rightarrow m = 4 - 2n$$

Sub into product equation:

$$(4 - 2n)^2 - n(4 - 2n) - 2n^2 = \frac{7}{2}$$

Solve and simplify to find:

$$\frac{m}{n} = \boxed{3}$$